

Exploring Magnets

Class 6 Science · Chapter 4 · Study Notes

1. Introduction

A magnet is an object that attracts certain materials, such as iron, towards itself. Magnets are all around us. They are used in refrigerator doors, in pencil boxes, in toys, in compasses and even in the speakers of a radio. Children often first meet a magnet as a toy that can pull paper clips or pins across a table without touching them.

The word "magnet" is believed to come from Magnesia, an ancient region in Greece. According to an old story, a shepherd named Magnes found that the iron tip of his stick and the nails in his shoes were pulled towards certain black stones lying on the ground. These stones were natural magnets. Today they are called lodestones or magnetite. The word "lodestone" means "leading stone", because such a stone, when hung freely from a thread, always points in the same direction.

Natural magnets are rare and are not very strong. Most of the magnets we use today are made by people. They are called artificial magnets. They are usually made from iron, steel or from alloys, which are mixtures of metals such as iron, nickel and cobalt.

In these notes we shall learn which materials a magnet can attract, what the poles of a magnet are, how magnets attract and repel each other, how a magnet helps us find directions, how to make a magnet, how to store magnets safely and where magnets are used in daily life.

2. Magnetic and Non-magnetic Materials

Materials that are attracted by a magnet are called magnetic materials. Materials that are not attracted by a magnet are called non-magnetic materials.

Common magnetic materials include:

- iron, for example a nail or an iron pin
- steel, for example a steel spoon or a paper clip
- nickel, for example a nickel coin in some countries
- cobalt, which is used in making strong magnets

Common non-magnetic materials include:

- paper and cardboard
- wood and plastic
- rubber and cloth
- glass
- copper, aluminium, gold and silver
- water and air

It is a very common mistake to think that every metal is attracted by a magnet. This is not true. Copper, aluminium, gold and silver are metals, but a magnet does not attract them. Only a few metals, mainly iron, nickel and cobalt, are magnetic.

Activity: sorting materials with a magnet. Collect objects such as a coin, a key, an eraser, a wooden ruler, a plastic comb, a paper clip, an iron nail, a glass marble, an aluminium foil piece and a piece of cloth. Bring a magnet close to each object one at a time. Place the objects that stick to the magnet in one group and the ones that do not in another group. Record the results in a table with two columns, "Magnetic" and "Non-magnetic". You will find that only the objects containing iron, nickel or cobalt are attracted.

A magnet can attract a magnetic object even through a thin layer of a non-magnetic material. For example, a magnet held under a sheet of paper or a thin wooden board can move an iron pin lying on top of it. However, if the layer is very thick, the magnet cannot pull the object because its force becomes weaker with distance.

3. Properties of a Magnet

A magnet has several important properties.

- A magnet attracts magnetic materials such as iron filings.
- A magnet has two ends called poles. The attraction is strongest at the poles and weakest in the middle.
- Like poles repel each other and unlike poles attract each other.
- A freely suspended magnet always comes to rest pointing in the north-south direction.
- A magnet loses its strength if it is heated strongly, hammered or dropped many times.

4. Poles of a Magnet

Every magnet has two poles. To see this, roll a bar magnet in a heap of iron filings and lift it up. You will see that a large number of filings stick near the two ends of the magnet, and very few stick near the middle. The places where the attraction is strongest are called the poles of the magnet.

The two poles of a magnet are named the north pole and the south pole. A bar magnet is often painted in two colours, or marked with the letters N and S, so that the poles can be identified easily.

The name "north pole" comes from the direction the pole points to. When a bar magnet is hung freely from a thread, one end always points towards the geographic north. This end is called the north-seeking pole, or simply the north pole. The other end, which points towards the south, is called the south-seeking pole, or the south pole.

Magnets come in different shapes and sizes. Some common shapes are:

- bar magnet, which is a long rectangular magnet
- horseshoe magnet, which is bent in the shape of the letter U
- ring magnet, which is shaped like a ring or a doughnut
- cylindrical magnet, which is a round rod
- disc magnet, which is a flat round magnet, like a coin
- magnetic needle, which is a thin and light magnet used in compasses

No matter what its shape is, every magnet has two poles. A magnet with only one pole has never been found. If you cut a bar magnet into two pieces, each piece becomes a smaller magnet with its own north pole and south pole. If you cut it again, the same thing happens. The two poles of a magnet cannot be separated.

5. Attraction and Repulsion

Take two bar magnets. Bring the north pole of one magnet close to the south pole of the other magnet. You will feel the magnets pull towards each other. Now bring the north pole of one magnet close to the north pole of the other. This time you will feel the magnets push each other away. The same happens if you bring two south poles close together.

This experiment shows the rule of magnetic poles:

- Unlike poles attract each other. A north pole attracts a south pole.

- Like poles repel each other. A north pole repels a north pole, and a south pole repels a south pole.

Repulsion means pushing away. Attraction means pulling closer.

How can you check whether a piece of iron is a magnet or not? A magnet attracts a piece of iron, and an ordinary piece of iron is also attracted by a magnet. So attraction alone does not prove that an object is a magnet. Repulsion is the sure test of magnetism. If an object is pushed away by one pole of a known magnet and pulled by the other pole, then the object is a magnet. An ordinary piece of iron is attracted by both poles of a magnet and is never repelled.

Bring a magnet slowly near a small toy car that has a magnet fixed on it. If you bring the like pole near, the car moves away, and if you bring the unlike pole near, the car moves towards the magnet. This is a good way to see attraction and repulsion in action. Some children even make two toy cars move without touching them, by using magnets.

6. Finding Directions with a Magnet

Long ago, sailors travelling on the sea had no maps of the oceans and no way of telling which direction their ship was going. They discovered that a piece of lodestone hung from a thread always settled down pointing in the north-south direction. They used this to find directions, even when the sky was cloudy and the Sun and stars could not be seen.

Today we use an instrument called a magnetic compass to find directions. A compass is a small round box with a glass top. Inside the box a light magnetic needle is balanced on a pivot so that it can rotate freely. The needle is usually coloured red or has an arrow at one end, and this end is the north-seeking pole. A dial marked with the directions north, south, east and west is fixed below the needle.

To use a compass, place it on a flat surface, away from other magnets and iron objects. Wait until the needle stops moving. The end of the needle that is marked N then points towards the north. Now turn the dial until the letter N on it lies below the needle. The other directions can then be read off from the dial.

Activity: making a simple direction finder. Tie a thread around the middle of a bar magnet and hang it from a stand so that it can move freely. When it settles, note which end is pointing north. Now push it gently and let it settle again. You will find that it always comes back to the same direction. Mark the north-pointing end with

red paint.

A compass is not reliable near iron objects, electric wires or other magnets, because they can pull the needle away from the north.

7. Magnetic Field

The space around a magnet in which the force of the magnet can be felt is called the magnetic field of the magnet. The magnetic field is stronger near the poles and becomes weaker as we move away.

A magnetic field cannot be seen, but we can see its effect. Place a sheet of paper over a bar magnet and sprinkle iron filings evenly on the paper. Tap the paper gently. The filings arrange themselves in curved lines that join the two poles of the magnet. These lines show the shape of the magnetic field. Outside the magnet, the field lines go from the north pole to the south pole.

The field lines are closer together near the poles, where the magnet is strongest, and farther apart in the region where the magnet is weak. If two unlike poles are placed near each other, the field lines join the two magnets. If two like poles are placed near each other, the lines bend away from each other.

8. The Earth is a Big Magnet

The Earth itself behaves like a giant bar magnet. This is why a freely suspended magnet or a compass needle always points in the north-south direction. The Earth's magnetic south pole lies near the geographic north and its magnetic north pole lies near the geographic south. This is the reason why the north-seeking pole of a compass needle points to the geographic north: unlike poles attract each other.

Some animals, such as certain birds, turtles and fish, are believed to sense the Earth's magnetic field. It may help them find their way when they travel long distances.

9. Making a Magnet

A piece of iron or steel can be turned into a magnet. One simple method is the single touch method.

- Take a steel needle or a long iron nail and place it on a table.
- Take a strong bar magnet and place one of its poles at one end of the needle.

- Stroke the needle from one end to the other in one direction only, using the same pole of the magnet.
- Lift the magnet well above the needle at the end of each stroke and bring it back to the starting end. Do not rub back and forth.
- Repeat the stroking about thirty to forty times.
- Test the needle by bringing it near some iron pins or paper clips. If it picks them up, it has become a magnet.

Steel keeps its magnetism for a long time, so steel is used for making permanent magnets. Soft iron loses its magnetism very quickly, so it is used to make temporary magnets. Electricity can also be used to make magnets, and these are called electromagnets. An electromagnet works only while an electric current flows through it and is used in cranes that lift heavy scrap iron in factories and junkyards.

10. Keeping Magnets Safe

Magnets can lose their strength if they are not stored carefully. The following precautions should be taken.

- Do not heat magnets, drop them or hit them with a hammer.
- Keep bar magnets in pairs with their unlike poles on the same side. Place a piece of wood between them, and put a piece of soft iron called a keeper across the ends.
- Keep horseshoe magnets with a piece of soft iron across their poles.
- Keep magnets away from mobile phones, televisions, computers, credit cards, memory cards and watches, because a strong magnet can damage them.
- Do not keep magnets near a compass, because it can make the needle point in the wrong direction.

11. Uses of Magnets

Magnets are useful in many ways.

- Compasses use a magnetic needle to show directions.
- Refrigerator doors have a magnetic strip which holds the door tightly shut.
- Pencil boxes and cupboards often have magnetic catches.
- Toys such as magnetic fishing games, puzzle boards and building blocks use magnets.

- Loudspeakers, earphones and electric motors contain magnets.
- Cranes with powerful electromagnets separate and lift iron and steel scrap.
- Some doctors use magnets to remove tiny iron splinters from the eye.
- Magnetic levitation trains, called maglev trains, float above the track and move very fast because like poles of magnets repel each other.
- Magnets are used in machines called MRI scanners, which help doctors look inside the body.

12. Electromagnets

An electromagnet is a magnet that works with the help of electricity. To make a simple electromagnet, wind a long insulated copper wire around an iron nail to form a coil, and connect the ends of the wire to a battery. When the current flows through the wire, the nail becomes a magnet and can pick up pins and paper clips. When the battery is disconnected, the nail loses its magnetism almost at once.

The strength of an electromagnet can be changed. It becomes stronger when more turns of wire are wound around the iron core, and when a bigger current flows through the coil. A permanent magnet cannot be switched off, but an electromagnet can be switched on and off whenever we like, and this is why it is so useful.

Electromagnets are used in electric bells, in cranes that lift heavy iron sheets and scrap in factories, in electric motors, and in many machines at home such as fans and mixers. A crane with a large electromagnet picks up a heap of scrap iron when the current is switched on, carries it across the yard, and drops it when the current is switched off.

13. More Activities and Experiments

Activity 1: the dancing paper clip. Place a paper clip on a sheet of thin cardboard. Move a magnet slowly under the cardboard. The paper clip follows the magnet, because the magnetic force acts through the cardboard.

Activity 2: the magnetic boat. Make a small paper boat and place an iron pin or a paper clip inside it. Float it in a shallow tray of water. Move a magnet along the outside of the tray, and the boat sails along with it. This shows that the magnetic force can act through glass, plastic and water.

Activity 3: finding the strongest part of a magnet. Dip a bar magnet into a tray of pins or paper clips, first at one end, then in the middle and then at the other end.

Count the number of pins sticking at each place. The largest number stick to the ends, which shows that the poles are the strongest parts of a magnet.

Activity 4: comparing the strength of two magnets. Use two different magnets and count how many paper clips each one can hold in a chain. The magnet that holds more clips is the stronger magnet. A bigger magnet is often, but not always, stronger.

Activity 5: making a floating compass. Magnetise a sewing needle by stroking it with a bar magnet. Push the needle through a small piece of cork or thermocol and float it in a bowl of water. Keep the bowl away from other magnets and iron objects. The needle will slowly turn until one end points to the north and the other end points to the south.

Activity 6: a magnet race. Fix a bar magnet on a toy car with its north pole facing forward. Hold another magnet in your hand with its north pole facing the car and move it slowly towards the car. The car runs away from you as like poles repel. Turn the magnet in your hand around so that its south pole faces the car. The car will now follow the magnet.

14. Common Misconceptions

- Misconception: all metals are attracted by a magnet. In fact only iron, nickel and cobalt and materials made from them are magnetic. Aluminium, copper, silver and gold are not attracted.
- Misconception: a magnet can attract every kind of object. A magnet does not attract wood, paper, plastic, glass, rubber or cloth.
- Misconception: if a magnet is cut, we get a piece with only a north pole and a piece with only a south pole. In fact each piece becomes a complete magnet with two poles.
- Misconception: a magnet always attracts another magnet. Two magnets attract only when unlike poles face each other. Like poles repel.
- Misconception: the north pole of a compass needle points to the Earth's magnetic north pole. The needle points towards the geographic north, and this end of the needle is attracted to the region near the Earth's geographic north, which is really a magnetic south pole.
- Misconception: a magnet stays strong forever, however it is treated. Heating, hammering and dropping can weaken a magnet, and storing magnets carelessly also reduces their strength over time.

15. The Story of Magnets

- About 600 BC, the Greek thinker Thales of Miletus wrote about the stones that attracted iron, which we now call lodestones.
- The Chinese discovered that a lodestone floating on water or hanging from a thread always pointed in the same direction, and by about the eleventh century magnetic compasses were being used by sailors to cross the seas.
- In 1600, an English scientist called William Gilbert published a book in which he showed that the Earth itself behaves like a giant magnet.
- In 1820, the Danish scientist Hans Christian Oersted found that an electric current flowing through a wire can move a compass needle. This discovery linked electricity with magnetism and led to the invention of electromagnets and electric motors.

16. Magnets in Everyday Objects

A refrigerator magnet is made of a soft, flexible strip of rubber mixed with tiny magnetic particles. It sticks to the steel door of the refrigerator and holds notes or pictures. A magnetic strip fixed along the edge of the refrigerator door pulls the door tight against the body, so that cold air does not escape.

A pencil box often has a small magnet fixed in its lid and a piece of iron or steel in the body. When the lid is closed, the magnet holds it shut. Many cupboard doors have similar catches.

The dark strip on the back of some bank cards is made of tiny magnetic particles that store information. A strong magnet brought near such a card can spoil the information stored on it, which is why we keep cards away from magnets.

17. Key Terms

- Magnet: an object that attracts iron and some other materials.
- Magnetic material: a material that is attracted by a magnet.
- Non-magnetic material: a material that is not attracted by a magnet.
- Pole: the end of a magnet where the attraction is strongest.
- North pole: the end of a freely suspended magnet that points north.
- South pole: the end of a freely suspended magnet that points south.
- Attraction: pulling towards each other.

- Repulsion: pushing away from each other.
- Compass: an instrument with a magnetic needle used to find directions.
- Magnetic field: the space around a magnet where its force can be felt.
- Lodestone: a natural magnet.
- Keeper: a piece of soft iron used to keep magnets from losing strength.

18. Quick Recap

- Magnets attract iron, steel, nickel and cobalt. They do not attract materials like wood, plastic, glass, copper or aluminium.
- Every magnet has two poles, a north pole and a south pole, and the two poles cannot be separated.
- Like poles repel and unlike poles attract. Repulsion is the sure test of a magnet.
- A freely suspended magnet points north-south. A compass uses this property to find directions.
- The Earth acts like a giant magnet.
- Magnets can be made by stroking iron or steel with a magnet, and they must be stored carefully.
- Magnets have many uses in homes, industry, transport and medicine.